

LS3 WiFi 2.4G/BLE

Product Specification

802.11a/b/g/n WiFi & BLE 5.4 Module 1T1R

(ESP32-S3R8)

Version V1.0

History

Document Release	Date	Modification	Initials	Approved
Version V1.0	2026/02/26	First Version		

1 Overview

ESP32-S3 is a low-power MCU-based system on a chip (SoC) with integrated 2.4 GHz Wi-Fi and Bluetooth® Low Energy (Bluetooth LE). It consists of high-performance dual-core microprocessor (Xtensa® 32-bit LX7), a ULP coprocessor, a Wi-Fi baseband, a Bluetooth LE baseband, RF module, and numerous peripherals.

1.1 Wi-Fi Features

- Complies with IEEE 802.11b/g/n
- Supports 20 MHz and 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Wi-Fi Multimedia (WMM)
- TX/RX A-MPDU, TX/RX A-MSDU
- Immediate Block ACK
- Fragmentation and defragmentation
- Automatic Beacon monitoring (hardware TSF)
- Four virtual Wi-Fi interfaces
- Simultaneous support for Infrastructure BSS in Station, SoftAP, or Station + SoftAP modes
Note that when ESP32-S3 scans in Station mode, the SoftAP channel will change along with the Station channel
- Antenna diversity
- 802.11mc FTM

1.2 Bluetooth

- Bluetooth LE: Bluetooth 5, Bluetooth Mesh
- High-power mode with up to 20 dBm transmission power
- Speed: 125 Kbps, 500 Kbps, 1 Mbps, 2 Mbps
- LE Advertising Extensions
- Multiple Advertising Sets
- LE Channel Selection Algorithm #2
- Internal co-existence mechanism between Wi-Fi and Bluetooth to share the same antenna

1.3 CPU and Memory

- Xtensa® dual-core 32-bit LX7 microprocessor
- Clock speed: up to 240 MHz
- CoreMark® score:
 - Two cores at 240 MHz: 1329.92 CoreMark; 5.54 CoreMark/MHz
- Five-stage pipeline
- 128-bit data bus and dedicated SIMD instructions

- Single precision floating point unit (FPU)
- Ultra-Low-Power (ULP) coprocessors:
 - ULP-RISC-V coprocessor
 - ULP-FSM coprocessor
- General DMA controller, with 5 transmit channels and 5 receive channels
- L1 cache
- ROM: 384 KB
- SRAM: 512 KB
- SRAM in RTC: 16 KB
- 4096-bit eFuse memory, up to 1792 bits for users
- Supported SPI protocols: SPI, Dual SPI, Quad SPI, Octal SPI, QPI and OPI interfaces that allow connection to flash, external RAM, and other SPI devices
- Flash controller with cache is supported
- Flash in-Circuit Programming (ICP) is supported

1.4 Peripherals

- 45 programmable GPIOs
 - 4 strapping GPIOs
 - GPIOs allocated for in-package memory:
 - * 6 GPIOs for either in-package flash or PSRAM
 - * 7 GPIOs when both in-package flash and PSRAM are integrated
- Connectivity interfaces:
 - Three UART interfaces
 - Two I2C interfaces
 - Two I2S interfaces
 - LCD interface
 - 8-bit ~ 16-bit DVP camera interface
 - Two SPI ports for communication with flash and RAM
 - Two general-purpose SPI ports
 - TWAI® controller, compatible with ISO 11898-1 (CAN Specification 2.0)
 - Full-speed USB OTG
 - USB Serial/JTAG controller
 - SD/MMC host controller with 2 slots
 - LED PWM controller, up to 8 channels
 - Two Motor Control PWM (MCPWM)
 - RMT (TX/RX)
 - Pulse count controller
- Analog signal processing:
 - Two 12-bit SAR ADCs, up to 20 channels

- Temperature sensor
- 14 capacitive touch sensing IOs
- Timers:
 - Four 54-bit general-purpose timers
 - 52-bit system timer
 - Three watchdog timers

1.5 Power Management

- Fine-resolution power control, including clock frequency, duty cycle, Wi-Fi operating modes, and individual internal component control
- Four power modes designed for typical scenarios: Active, Modem-sleep, Light-sleep, Deep-sleep
- Power consumption in Deep-sleep mode is 7 μ A
- RTC memory remains powered on in Deep-sleep mode

1.6 Security

- Secure boot - permission control on accessing internal and external memory
- Flash encryption - memory encryption and decryption
- Cryptographic hardware acceleration:
 - SHA Accelerator (FIPS PUB 180-4)
 - AES Accelerator (FIPS PUB 197)
 - RSA Accelerator
 - HMAC Accelerator
 - RSA Digital Signature Peripheral (RSA_DS)
 - Random Number Generator (RNG)

1.9 RF Module

- Antenna switches, RF balun, power amplifier, low-noise receive amplifier
- Up to +21 dBm of power for an 802.11b transmission
- Up to +19.5 dBm of power for an 802.11n transmission
- Up to -104.5 dBm of sensitivity for Bluetooth LE receiver (125 Kbps)

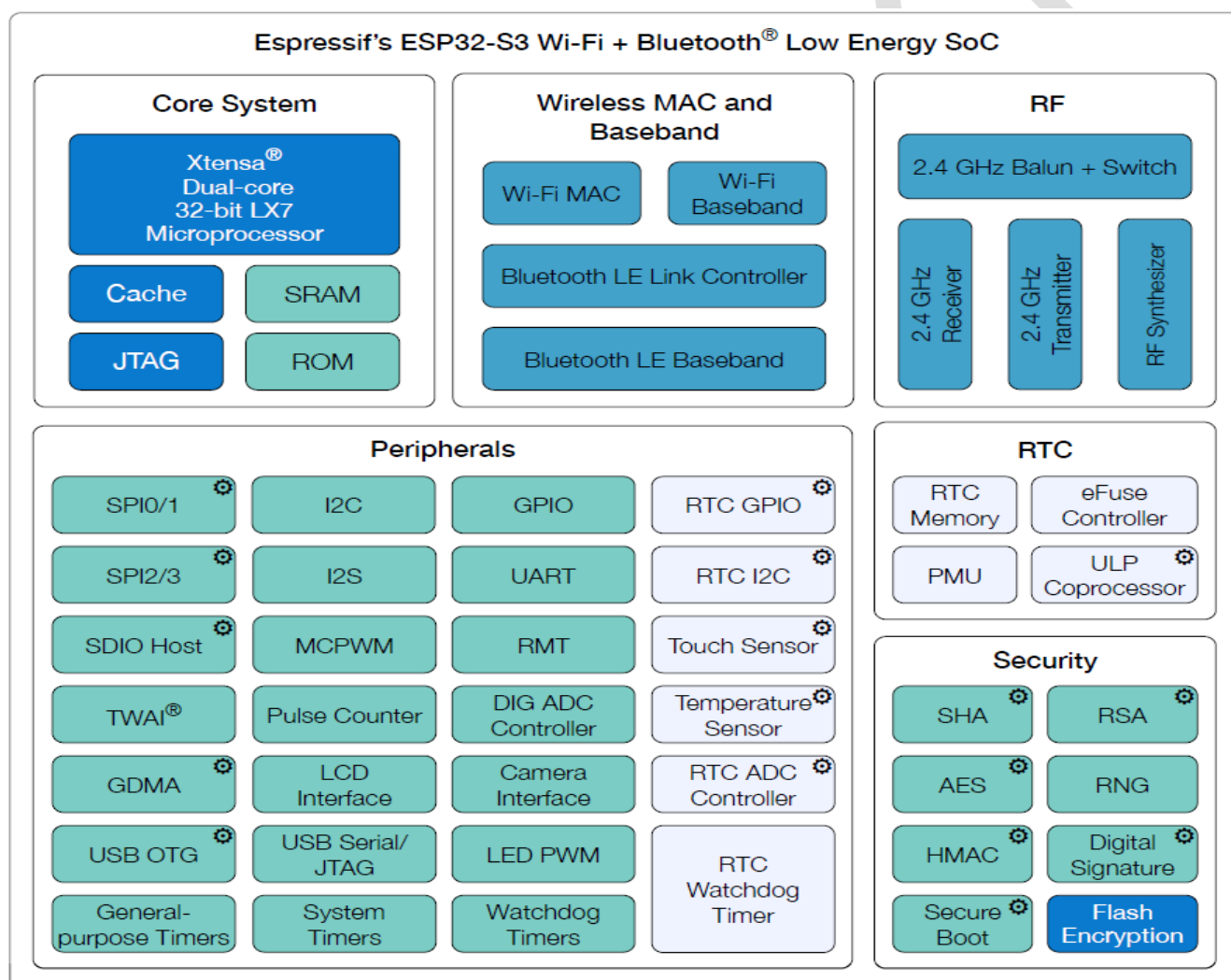
1.9 Applications

With low power consumption, ESP32-S3 is an ideal choice for IoT devices in the following areas:

- Smart Home
- Industrial Automation
- Health Care
- Consumer Electronics
- Smart Agriculture

- POS Machines
- Service Robot
- Audio Devices
- Generic Low-power IoT Sensor Hubs
- Generic Low-power IoT Data Loggers
- Cameras for Video Streaming
- USB Devices
- Speech Recognition
- Image Recognition
- Wi-Fi + Bluetooth Networking Card
- Touch and Proximity Sensing

2 Application Diagram

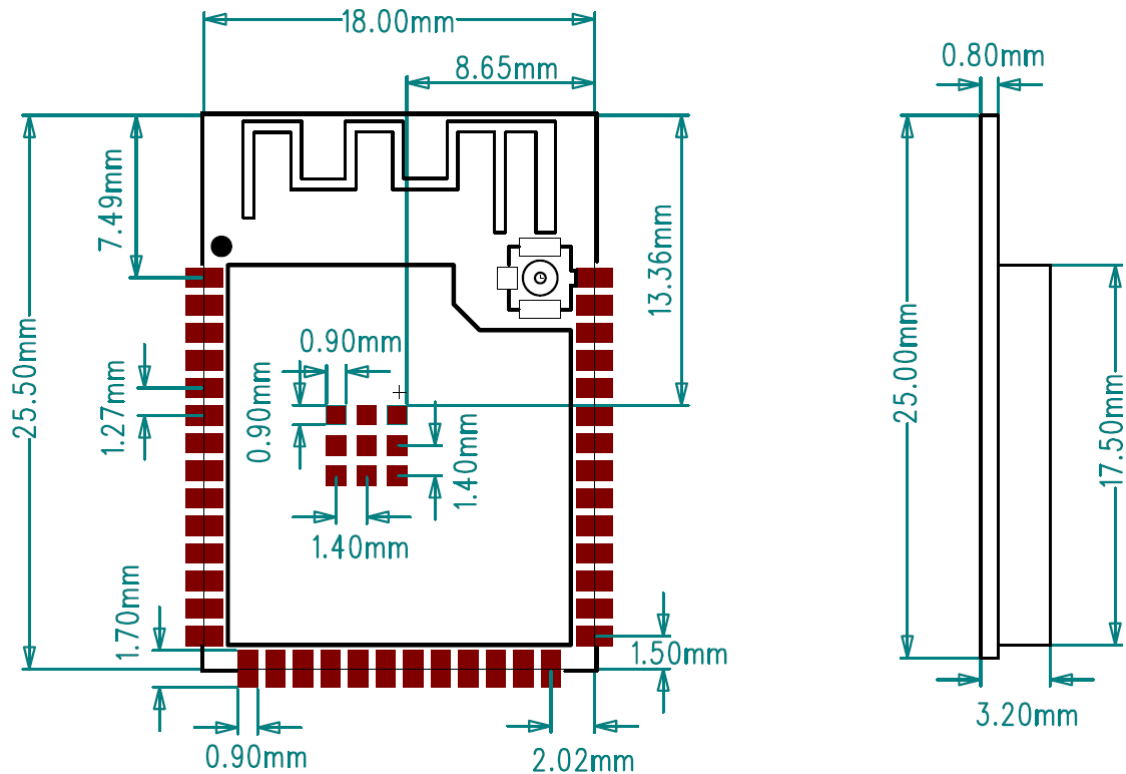


3 Recommended Operating Conditions

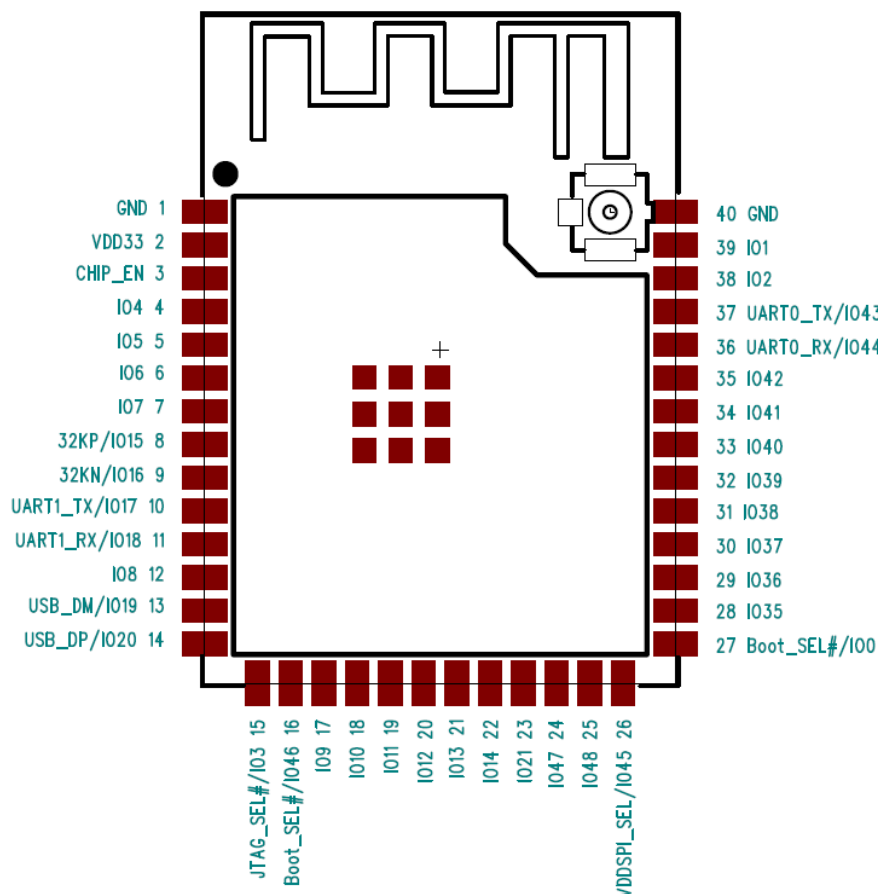
Parameter	Min	Typ	Max	Unit
Operation Voltage	3.0	3.3	3.6	V
Working Current	600	-	-	mA
Operation Temperature	-40	-	65	°C

4 Dimensions (Units: mm) (Dimensional tolerance $\pm 10\%$)

Note: The height of the module with a shielding cover is $2.2 \pm 0.2 \text{mm}$, while the overall height of the module without a shielding cover is $1.8 \pm 0.2 \text{mm}$



5. Pin Definition



There are 9 ground pads at the bottom of the module, which are used to enhance grounding and heat dissipation.

6 Pin Description

Pin	Function	Type	Voltage	Description
1	GND	GND	-	GND
2	VDD33	IPMU	3.3V	3.3V Power Supply
3	CHIP_EN	I	3.3V	Chip power-on enable, There is no pull-up inside the module
4	IO4	I/O	3.3V	RTC_GPIO4, GPIO4, TOUCH4, ADC1_CH3
5	IO5	IPMU	3.3V	RTC_GPIO5, GPIO5, TOUCH5, ADC1_CH4
6	IO6	I/O	3.3V	RTC_GPIO6, GPIO6, TOUCH6, ADC1_CH5
7	IO7	I/O	3.3V	RTC_GPIO7, GPIO7, TOUCH7, ADC1_CH6
8	32KP/IO15	I/O	3.3V	RTC_GPIO15, GPIO15, U0RTS, ADC2_CH4, XTAL_32K_P
9	32KN/IO16	I/O	3.3V	RTC_GPIO16, GPIO16, U0CTS, ADC2_CH5, XTAL_32K_N
10	UART1_TX/IO17	IPMU	3.3V	RTC_GPIO17, GPIO17, U1TXD, ADC2_CH6
11	UART1_RX/IO18	I/O	3.3V	RTC_GPIO18, GPIO18, U1RXD, ADC2_CH7, CLK_OUT3
12	IO8	I/O	3.3V	RTC_GPIO8, GPIO8, TOUCH8, ADC1_CH7, SUBSPICS1
13	USB_DM/IO19	I/O	3.3V	RTC_GPIO19, GPIO19, U1RTS, ADC2_CH8, CLK_OUT2, USB_D-
14	USB_DP/IO20	I/O	3.3V	RTC_GPIO20, GPIO20, U1CTS, ADC2_CH9, CLK_OUT1, USB_D+
15	JTAG_SEL#/IO3	I/O	3.3V	RTC_GPIO3, GPIO3, TOUCH3, ADC1_CH2
16	BOOT_SEL#/IO46	I/O	3.3V	GPIO46
17	IO9	I/O	3.3V	RTC_GPIO9, GPIO9, TOUCH9, ADC1_CH8, FSPIHD, SUBSPIHD
18	IO10	I/O	3.3V	RTC_GPIO10, GPIO10, TOUCH10, ADC1_CH9, FSPICS0, FSPIIO4, SUBSPICS0
19	IO11	I/O	3.3V	RTC_GPIO11, GPIO11, TOUCH11, ADC2_CH0, FSPID, FSPIIO5, SUBSPID
20	IO12	I/O	3.3V	RTC_GPIO12, GPIO12, TOUCH12, ADC2_CH1, FSPICLK, FSPIIO6, SUBSPICLK
21	IO13	I/O	3.3V	RTC_GPIO13, GPIO13, TOUCH13, ADC2_CH2, FSPIQ, FSPIIO7, SUBSPIQ
22	IO14	I/O	3.3V	RTC_GPIO14, GPIO14, TOUCH14, ADC2_CH3, FSPIWP, FSPIDQS, SUBSPIWP
23	IO21	I/O	3.3V	RTC_GPIO21, GPIO21
24	IO47	I/O	3.3V	SPICLK_P_DIFF, GPIO47, SUBSPICLK_P_DIFF
25	IO48	I/O	3.3V	SPICLK_N_DIFF, GPIO48, SUBSPICLK_N_DIFF
26	VDDSPI_SEL/IO45	I/O	3.3V	GPIO45
27	BOOT_SEL#/IO0	I/O	3.3V	RTC_GPIO0, GPIO0
28	IO35	I/O	3.3V	SPIIO6, GPIO35, FSPID, SUBSPID
29	IO36	I/O	3.3V	SPIIO7, GPIO36, FSPICLK, SUBSPICLK
30	IO37	I/O	3.3V	SPIDQS, GPIO37, FSPIQ, SUBSPIQ
31	IO38	I/O	3.3V	GPIO38, FSPIWP, SUBSPIWP
32	IO39	I/O	3.3V	MTCK, GPIO39, CLK_OUT3, SUBSPICS1
33	IO40	I/O	3.3V	MTDO, GPIO40, CLK_OUT2
34	IO41	I/O	3.3V	MTDI, GPIO41, CLK_OUT1
35	IO42	I/O	3.3V	MTMS, GPIO42
36	UART0_RX/IO44	I/O	3.3V	U0RXD, GPIO44, CLK_OUT2
37	UART0_TX/IO43	I/O	3.3V	U0TXD, GPIO43, CLK_OUT1
38	IO2	I/O	3.3V	RTC_GPIO2, GPIO2, TOUCH2, ADC1_CH1
39	IO1	I/O	3.3V	RTC_GPIO1, GPIO1, TOUCH1, ADC1_CH0
40	GND	GND	-	GND
41~49	GND	GND	-	GND

7 Current consumption

Parameter	Test Item	TX Power	Current	Unit
WiFi TX	2.4G,11b, CCK,1Mbps	21dBm	315	mA
	2.4G,11b, CCK,11Mbps	21dBm	212	mA
	2.4G,11g, OFDM, 6Mbps	20dBm	320	mA
	2.4G,11g, OFDM, 54Mbps	18dBm	273	mA
	2.4G,11n,HT20, MCS0	19dBm	299	mA
	2.4G,11n,HT20, MCS7	17.5dBm	267	mA
	2.4G,11n,HT40, MCS0	18.5dBm	305	mA
	2.4G,11n,HT40, MCS7	17.5dBm	263	mA
Stop WiFi TX	-	-	99	mA
WiFi RX	-	-	102	mA
Stop WiFi RX	-	-	96	mA
BLE TX	BLE 1M	21.5dBm	362	mA
Stop BLE TX	-	-	98	mA
BLE RX	-	-	109	mA
Stop BLE RX	-	-	96	mA

8 RF Characteristics

8.1 WiFi 2.4G Transmitter Characteristics

Parameter	Test Item	Typical Value	CH1	CH7	CH13	Unit
Output Power	11b,CCK, 1Mbps	21±2dBm,EVM<-15dB	21.03	21.08	20.79	dBm
	11b,CCK,11Mbps	21±2dBm,EVM<-15dB	21.25	21.34	20.96	dBm
	11g,OFDM, 6Mbps	20±2dBm,EVM<-15dB	19.95	19.91	19.63	dBm
	11g,OFDM,54Mbps	18±2dBm,EVM<-25dB	18.22	18.17	17.86	dBm
	11n, HT20 MCS0	19±2dBm,EVM<-15dB	19.09	19.19	18.76	dBm
	11n, HT20 MCS7	17.5±2dBm,EVM<-27dB	17.60	17.78	17.50	dBm
Parameter	Test Item	Typical Value	CH3	CH7	CH11	Unit
Output Power	11n, HT40 MCS0	18.5±2dBm,EVM<-15dB	18.59	18.67	18.73	dBm
	11n, HT40 MCS7	17.5±2dBm,EVM<-27dB	17.58	17.62	17.72	dBm

8.2 WiFi 2.4G Receiver Characteristic

Parameters	Test Item	CH1	CH7	CH13	Unit
2.4G Receive Sensitivity	11b, CCK, 1M , <-76dBm@8%PER	-98	-98	-98	dBm
	11b, CCK, 11M , <-76dBm@8%PER	-88	-88	-88	dBm
	11g, OFDM, 6M , <-82dBm@10%PER	-91	-91	-91	dBm
	11g, OFDM, 54M , <-65dBm@10%PER	-75	-75	-75	dBm
	11n, HT20 MCS0, <-82dBm@10%PER	-91	-91	-91	dBm
	11n, HT20 MCS7, <-64dBm@10%PER	-73	-73	-73	dBm
Parameters	Test Item	CH3	CH7	CH11	Unit
2.4G Receive Sensitivity	11n, HT40 MCS0, <-79dBm@10%PER	-89	-89	-89	dBm
	11n, HT40 MCS7, <-61dBm@10%PER	-70	-70	-70	dBm

8.3 BLE TX Performance

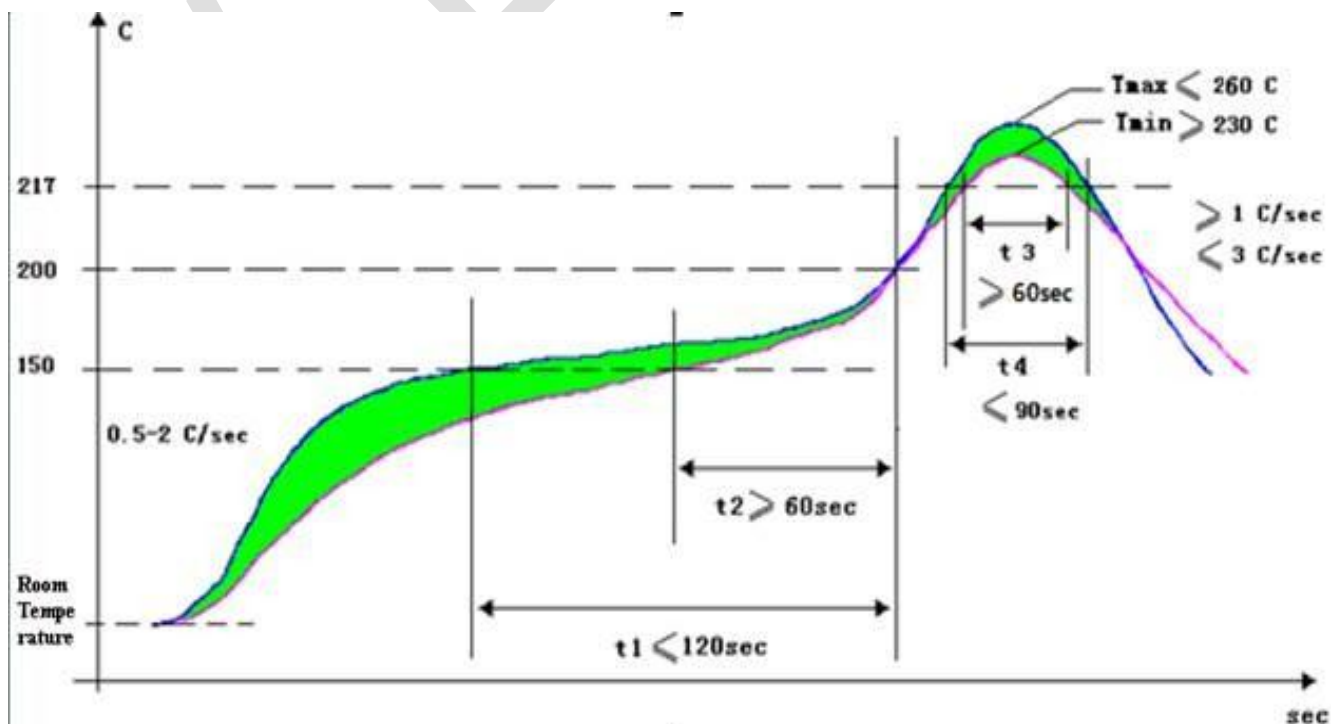
Parameter	Test Item	Typical Value	Channel			Unit
			CH0	CH20	CH39	
Output power	BLE_1M	21.5±2	21.6	22.07	22.02	dBm
Output power	BLE_2M	21.5±2	21.71	22.25	22.26	dBm

8.4 BLE RX Performance

Parameter	Test Item	Typical Value	Channel			Unit
			CH0	CH20	CH39	
Sensitivity PER>30%	BLE_1M	<-90	-95	-95	-95	dBm
Sensitivity PER>30%	BLE_2M	<-87	-90	-90	-90	dBm

9 Parameter Requirements on Lead-Free Reflow Soldering

- shows the thermal profile of lead-free reflow soldering



Parameters of lead-free reflow soldering

Zone	Time	Heating Rate	Peak Temperature	Cooling Rate
Preheat zone(40~150℃)	60~150s	≤2.0℃/s	-	-
Soak zone(150~200℃)	60~120s	<1.0℃/s	-	-
Reflow zone(>217℃)	60~90s	-	230~260℃	-
Cooling zone(Tmax~180℃)	-	-	-	1.0℃/s≤Slope≤4.0℃/s

Note:

- Preheat zone: The temperate range is 40–150℃ (104–302°F), the heating rate must be about 2.0℃/s (36°F/s), and the zone duration must be 60–150s.
- Soak zone: The temperature range is 150–200℃ (302–392°F), the heating rate must be less than 1.0℃/s (34°F/s), and the zone duration must be 60–120s. Slow heating is required; otherwise, soldering is poor.
- Reflow zone: The zone temperature increases from 217℃ (423°F) to Tmax and then decreases to 217℃ (423°F) again. The zone duration must be 60–90s.
- Cooling zone: The zone temperature decreases from Tmax to 180℃ (356°F). The cooling rate cannot exceed 4.0℃/s (39°F/s).
- It should take no more than 6 minutes for the ambient temperature to increase from 25℃ (77°F) to 250℃ (482°F).
- The thermal profile shown in the preceding figure provides recommended values. Customers need to adjust the values based on actual production.
- Typically, the duration of the reflow zone is 60–90s. For the boards with great heat capacity, the duration can be prolonged to 120s. For details about the requirements on package thermal resistance, see the IPC/JEDEC J-STD-020D standard. For details about the method of measuring the package temperature, see the JEP 140 standard.

● Thermal resistance standard for the lead-free package

Package Thickness	Volume mm ³ <350	Volume mm ³ 350~2000	Volume mm ³ >2000
< 1.6mm	260℃	260℃	260℃
1.6mm ~ 2.5mm	260℃	250℃	245℃
> 2.5mm	250℃	245℃	245℃

The component soldering terminals (such as the solder balls and pins) and external heat sinks are not considered for volume calculation.

The method of measuring the reflow soldering thermal profile is as follows:

According to the JEP140 standard, to measure the package temperature, you are advised to place the temperature probe of the thermocouple close to the chip surface if the chip package is thin, or to drill a hole on the package surface and place the temperature probe of the thermocouple into the hole if the chip package is thick. The second method is recommended based on the thickness of most chip packages. However, this method is not applicable if the chip package is too thin to drill a hole.

10 Order Information

Module	Part number	Description
LS3	LS3-NI-NS	LS3 (a/b/g/n/ax) WiFi/BLE Module , No IPEX , No Shield cover
	LS3-NI-WS	LS3 (a/b/g/n/ax) WiFi/BLE Module , No IPEX , With Shield cover
	LS3-WI-NS	LS3 (a/b/g/n/ax) WiFi/BLE Module , With IPEX , No Shield cover
	LS3-WI-WS	LS3 (a/b/g/n/ax) WiFi/BLE Module , With IPEX , With Shield cover